

The 17th International Conference on Ambient Systems, Networks and Technologies (ANT)
April 14-16, 2026, Istanbul, Türkiye

EdgeSeal-FHE: A Privacy-Preserving Encryption Framework for End-to-End Encrypted Health Monitoring

Alex Jeter^{a,*}, Justice Pankey-Thompson^a, Sanaz Rahimi Moosavi^a, Ali Jalooli^a

^a*Department of Computer Science, California State University Dominguez Hills, Carson, CA 90747, USA*

Abstract

The increasing use of the Internet of Things (IoT) and wearable devices in healthcare exposes sensitive patient data to security risks when transmitted through untrusted networks. In this paper, we present EdgeSeal-FHE, an end-to-end encryption framework that applies Fully Homomorphic Encryption (FHE) at the edge to ensure that physiological data remain encrypted from generation to cloud analysis with no plaintext exposure. The system uses the Cheon-Kim-Kim-Song (CKKS) scheme via TenSEAL to support computation on encrypted medical embeddings, enabling continuous confidentiality and integrity without disrupting real-time processing. Evaluation results show that EdgeSeal-FHE encrypts 300-dimensional embeddings in 4.52 ms at 128-bit security, achieves end-to-end latency of 35.48 ms, and preserves 99.72% data quality with less than 0.5% reconstruction loss. Compared with state-of-the-art CKKS healthcare systems, our approach delivers higher throughput while remaining under the 100 ms real-time threshold across all device classes, including low-end smartphones. These results demonstrate that practical, real-time encrypted healthcare monitoring is achievable without sacrificing security, accuracy, or responsiveness.

© 2026 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer review under the responsibility of the scientific committee of the Program Chairs

Keywords: Fully Homomorphic Encryption (FHE); CKKS; Privacy-Preserving Machine Learning; IoT Healthcare; Edge Computing; Encrypted Analytics; Real-Time Monitoring

1. Introduction

The rapid expansion of wearable devices and the Internet of Things (IoT) enabled health platforms has enabled continuous collection of physiological signals such as ECG, blood pressure, heart rate, and oxygen saturation. These data streams support early detection of clinical deterioration, personalized interventions, and improved patient outcomes [8]. As remote monitoring becomes more common in chronic care, emergency response, and postoperative recovery, ensuring the security of these data pipelines is increasingly critical.

Healthcare IoT systems transmit sensitive information across networks, cloud services, and hospital infrastructures, where data are vulnerable to interception, manipulation, and unauthorized access. Compromised health data can result not only in privacy violations but also direct risks to patient safety, as adversarial modifications to vital signs may cause incorrect diagnoses or delayed treatment. Security breaches additionally erode patient trust in remote monitoring systems [15], while regulations such as the Health Insurance Portability and Accountability Act of 1996 (HIPAA) mandate strict protection of confidentiality, integrity, and availability for digital health platforms [14].

* Corresponding author.

E-mail address: ajeter2@csudh.edu

Current security approaches—including Transport Layer Security (TLS) for data in transit, Advanced Encryption Standard (AES) for data at rest [7], and blockchain-based access control [12]—provide only partial protection. These methods secure isolated stages of the data lifecycle but do not guarantee continuous confidentiality from the moment data are generated. Intermediate nodes such as edge gateways, cloud preprocessing services, and analytics engines often handle plaintext [13]. Additionally, wearables commonly lack strong security features due to constrained hardware, limited update mechanisms, or weak communication protocols. As a result, few existing systems achieve true end-to-end protection for healthcare IoT data [13].

To address these limitations, we propose *EdgeSeal-FHE*, an end-to-end privacy-preserving framework that applies Fully Homomorphic Encryption (FHE) at the edge, ensuring that no plaintext leaves the local device environment. FHE enables computation directly on encrypted data, allowing cloud analytics to operate without exposing sensitive information. In our design, physiological measurements are generated, transmitted to a trusted edge device over a trusted short-range network, then encrypted and remain protected throughout transmission, storage, and cloud-based computation. By integrating encryption and integrity validation throughout the pipeline, *EdgeSeal-FHE* prevents plaintext exposure at intermediate nodes and ensures that downstream analytics operate on tamper-resistant data.

The contributions of this work are threefold. First, we design a secure IoT-to-cloud architecture that applies the CKKS homomorphic encryption scheme, implemented with the TenSEAL library, to protect ECG, BP, SpO₂, and other physiological signals from acquisition to analysis. Second, we establish a threat model and formalize confidentiality and integrity guarantees under adversarial network conditions. Third, we enhance diagnostic reliability by ensuring that encrypted medical embeddings remain trustworthy and unaltered, allowing clinicians to rely on cloud-processed results without risking exposure of sensitive information.

The remainder of this paper is organized as follows. Section 2 reviews prior work in secure healthcare analytics. Section 3 introduces the system architecture and threat assumptions. Section 4 presents the *EdgeSeal-FHE* framework. Section 5 describes the evaluation setup. Section 6 evaluates real-time feasibility on resource-constrained devices. Section 7 concludes the paper.

2. Related Work

Elhoseny et al. [7] proposed a security model for medical data transmission in IoT-based healthcare systems that consists of four sequential processes. First, patient diagnostic data are encrypted using a hybrid AES–RSA scheme. Next, the resulting ciphertext is embedded into a medical cover image using either one-level or two-level 2D discrete wavelet transform (2D-DWT), producing a stego-image. The embedded data are subsequently extracted from the stego-image, and finally decrypted to recover the original medical information.

Cheon et al. [5] introduced a novel homomorphic encryption scheme for approximate arithmetic, originally called Homomorphic Encryption for Arithmetic of Approximated Numbers (HEAAN) and later renamed to Cheon-Kim-Kim-Song (CKKS) after the authors. Their scheme supports homomorphic addition and multiplication on encrypted messages by encoding plaintext vectors into ciphertext. All operations performed on that ciphertext introduce controlled noise resulting in approximate rather than exact decrypted outputs however, the error is sufficiently small for use in practical applications.

Benaissa et al. [2] introduced TenSEAL, a Privacy-Preserving Machine Learning (PPML) library based on FHE that enables encrypted tensor operations such as addition and multiplication for deep learning frameworks such as PyTorch and TensorFlow. Built on the CKKS scheme implemented in Microsoft SEAL, TenSEAL facilitates conversions from plaintext tensors to their encrypted counterparts. Their results on standard benchmarks such as MNIST demonstrate the growing practicality of FHE in PPML for real-world problems.

Lam et al. [10] proposed a framework that enables neural-network inference on data encrypted using FHE in a cloud environment. In their solution, sensitive data are encrypted via a FHE scheme by the client and then transmitted to the cloud, where a privacy-enhanced neural network performs inference on the ciphertext, producing encrypted outputs without exposing raw data. Their framework utilizes a dual network structure in which select components operate in plaintext, while the rest perform on ciphertexts. Their results demonstrate the practicality of their approach on many tasks, including facial recognition and text classification, achieving state-of-the-art accuracy and low computational cost.

In Private and Fast Text Classification (PrivFT) [1], Al et al. proposed a framework that provides 2 policy-preserving services: (1) Providing secure inference on ciphertexts encrypted by FHE by models pre-trained on plaintext, and (2) training models directly on FHE encrypted data to obtain models capable of operating on encrypted inputs. Their approach utilizes a GPU implementation of the CKKS homomorphic encryption scheme, achieving a performance speedup of 1 to 2 orders of magnitude when compared to CPU implementations for data inference and 5 orders of magnitude speedup for training on encrypted data.

3. System Model

As shown in Fig. 1, the system follows a multi-layer architecture designed to ensure that physiological data remain encrypted from acquisition to cloud-based analysis. The pipeline consists of five main components.

(1) **Wearable IoT devices** collect continuous physiological measurements (e.g., heart rate, temperature, SpO₂, motion signals) and transmit them to a nearby edge device over secure local communication.

(2) **Edge devices** act as trusted gateways responsible for lightweight preprocessing, including time alignment, noise reduction, and formatting. Critically, the edge device performs FHE immediately after preprocessing, ensuring that no raw sensor data leave the local environment.

(3) **FHE encryption module** applies the CKKS scheme to transform preprocessed sensor vectors into ciphertexts that support mathematical operations without revealing underlying values. All subsequent handling of data occurs on encrypted representations.

(4) **IoT cloud platform** receives encrypted data from multiple edge devices and performs homomorphic computations such as anomaly scoring, risk trend analysis, or aggregation across users. Since all values remain encrypted, the cloud infrastructure operates under an honest-but-curious model without access to plaintext.

(5) **Secure analytics layer** performs downstream processing entirely on ciphertext, enabling end-to-end encrypted inference and producing encrypted results that can only be decrypted by authorized healthcare recipients.

Trust Boundary and Assumptions. Plaintext exists only within the wearable device and the trusted edge node prior to encryption. After FHE encryption, all subsequent entities—including communication networks, cloud services, storage layers, and analytics engines—are considered untrusted. These components may handle ciphertext but cannot decrypt or infer the underlying sensor values. Thus, the trust boundary is strictly enforced at the edge, and all data beyond this point are assumed to be exposed to potential adversarial behavior.

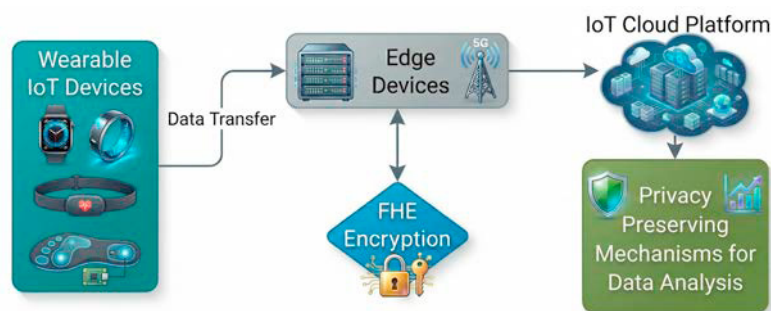


Fig. 1: System Model for Privacy-Preserving IoT Data Analysis

3.1. Data Flow and Security Life Cycle

The data life cycle begins when wearable IoT devices generate continuous physiological measurements and transmit them to a trusted edge node. The edge device performs lightweight preprocessing and then immediately applies CKKS-based FHE, ensuring that no plaintext leaves the local environment.

Once encrypted, the ciphertext is transferred to the cloud platform, where analytical or diagnostic operations are executed directly on encrypted data. Because all computations are performed homomorphically, the cloud never accesses or reconstructs plaintext values.

This pipeline enforces continuous end-to-end protection: plaintext exists only within the device–edge trust boundary, while all downstream processing, storage, and analytics operate exclusively on encrypted representations. This guarantees confidentiality and integrity throughout the data transmission and computation workflow.

Threat Model and Implementation. We assume an honest-but-curious cloud adversary that follows protocols correctly but attempts to infer plaintext from encrypted data. Wearable devices transmit raw physiological signals to the edge device (typically a smartphone, though our implementation is platform-agnostic) over short-range secure communication. The edge performs embedding generation and CKKS encryption with parameters tuned across security levels (polynomial degrees 4096/8192/16384), selecting degree 8192 with coefficient chain [60, 40, 40, 60] as optimal for balancing 128-bit security and latency. TenSEAL’s native vector batching encrypts full 300-dimensional embeddings in single operations. Analysis showed single-threaded processing achieves best performance (275.5 vectors/sec) due to Python’s Global Interpreter Lock preventing parallel FHE speedup.

4. Proposed Solution

EdgeSeal-FHE introduces a cryptographically enforced IoT-to-cloud pipeline that ensures physiological data remain encrypted throughout their entire lifecycle. Unlike conventional healthcare IoT architectures that expose plaintext during intermediate processing, our design shifts all encryption to the trusted edge device and performs analytics exclusively on ciphertext.

4.1. Design Goals

EdgeSeal-FHE is structured around three primary objectives: (1) *Continuous confidentiality*: ensure that raw sensor data are encrypted immediately after generation and never appear in plaintext beyond the device–edge trust boundary; (2) *Encrypted computation*: support mathematical operations on encrypted medical embeddings using the CKKS homomorphic encryption scheme; (3) *Real-time feasibility*: achieve end-to-end latency suitable for continuous health-care monitoring while preserving security guarantees.

4.2. Architecture Overview

The framework follows a layered architecture comprising IoT devices, trusted edge nodes, and an untrusted cloud. IoT devices forward raw measurements to an edge gateway, where lightweight preprocessing (e.g., normalization, timestamping) is performed. The edge then encrypts all vectors using CKKS before transmitting them to the cloud. The cloud stores ciphertext and executes homomorphic analytics—such as risk scoring, trend detection, or aggregation—without access to plaintext values. Only authorized recipients holding the private key may decrypt the final results.

4.3. Homomorphic Encryption Workflow

EdgeSeal-FHE employs the CKKS scheme via the TenSEAL library to support approximate arithmetic on real-valued medical signals. The encryption module operates under four parameter profiles selected based on performance and security requirements: (1) a *fast* configuration (poly modulus degree 4096) for low-latency applications, (2) a *balanced* configuration (8192) providing 128-bit security, (3) a *high-security* configuration (16384), and (4) a *high-precision* variant that extends coefficient moduli for improved numerical fidelity. After preprocessing, vectors are encoded and encrypted into CKKS ciphertexts and serialized for transmission. Integrity is verified using cryptographic hash functions such as SHA-256. Authorized parties possessing the private key may later decrypt the cloud-processed ciphertexts.

4.4. Edge Processing Mechanisms

The edge node is optimized for low-latency encryption and scalability. Multiple CKKS encryption contexts can be executed in parallel across threads to meet real-time throughput demands. Benchmark metrics include encryption latency, memory overhead, and effective throughput under different parameter profiles. Planned extensions include partial homomorphic inference at the edge to further reduce cloud dependency.

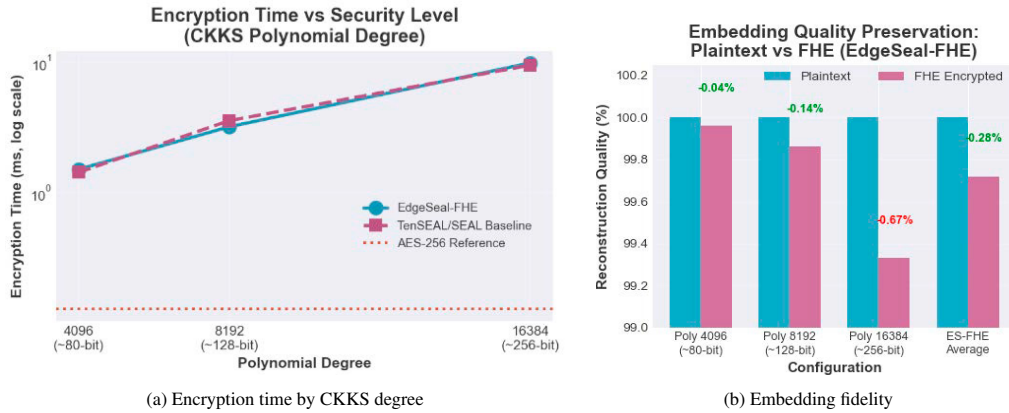


Fig. 2: EdgeSeal-FHE encryption latency and embedding quality.

4.5. Cloud Processing Mechanisms

The cloud is treated as honest-but-curious and performs computation solely on ciphertext. Supported operations include: (1) *Homomorphic linear scoring* using ciphertext–plaintext multiplication and encrypted summation; (2) *Encrypted aggregation* across multiple samples (ciphertext–ciphertext addition); and (3) *Thresholding via polynomial approximation* for decision boundaries. Because CKKS enables SIMD-style batching, the cloud can evaluate analytics on many encrypted vectors in parallel. Evaluation circuits are optimized to avoid exceeding noise budgets or the multiplicative depth supported by the chosen modulus chain.

5. Experimental Setup

We evaluate EdgeSeal-FHE on an Intel Core i7-10700K workstation (8 cores, 3.8–5.1 GHz), 32 GB RAM, and Windows 10/11. The system is implemented in TenSEAL v0.3.14 [2] using CKKS parameters from Microsoft SEAL [4]. Unless stated otherwise, we use a polynomial degree of 8192 (128-bit security), coefficient moduli {60, 40, 40, 60}, and scale 2^{40} . Galois keys are pre-generated for ciphertext rotations.

We evaluate both synthetic and real IoT health data, including 300-dimensional autoencoder embeddings, vital-sign triples (SpO_2 , HR, temperature), ECG sequences (300–500 samples), and 6–9 dimensional accelerometer/gyroscope vectors.

Metrics include encryption/decryption latency, ciphertext size, throughput, memory overhead, and Mean Squared Error (MSE) between plaintext and decrypted outputs. Real-time feasibility is defined as end-to-end latency < 100 ms [3].

Four experiment groups were conducted: (1) baseline overhead (CKKS vs. plaintext vs. AES-256 [6]); (2) comparison against multiple CKKS systems (2017–2025) [9, 2, 11, 5]; (3) sensitivity analysis of polynomial degrees 4096, 8192, 16384; (4) simulated performance on low-, mid-, and high-end smartphones. Each configuration was repeated across twenty trials with 95% confidence intervals.

5.1. Encryption Performance and Embedding Fidelity

Figure 2a shows encryption latency across CKKS polynomial degrees for EdgeSeal-FHE compared with the TenSEAL/SEAL baseline and an AES-256 reference. EdgeSeal-FHE achieves 4.52 ms at degree 8192 and 8.60 ms at degree 16384, closely tracking the TenSEAL/SEAL baseline while maintaining consistent performance across all security levels. Latency increases approximately linearly with polynomial modulus size, matching the expected CKKS complexity scaling.

Figure 2b demonstrates 99.72% reconstruction fidelity across all configurations. MSE ranges from 4.06×10^{-19} to 6.76×10^{-18} , confirming negligible numerical loss for medical embeddings. High fidelity is achieved through CKKS approximate arithmetic [5], balanced modulus selection, and L2-normalized inputs.

Table 1: End-to-End Performance: EdgeSeal-FHE vs. Literature Best

Metric	EdgeSeal-FHE	Literature Best
Encryption Time (8192 poly)	4.52 ms	2.5 ms [11]
Ciphertext Size (300-dim)	326.4 KB	~300 KB (avg.)
Throughput	179 samples/s	0.4 samples/s [11]
Accuracy Loss	< 0.5%	0–2% (varies)
Real-time (< 100 ms)	✓ Yes	55 No (most)

Table 2: Smartphone Hardware Profiles Used in Edge Simulation

Specification	Low-end	Mid-range	High-end
RAM	2 GB	4 GB	8 GB
CPU Cores	4	8	8
CPU Frequency	1.5 GHz	2.0 GHz	3.0 GHz
Perf. Scale	50%	70%	100%

5.2. End-to-End Performance

Table 1 summarizes end-to-end results. EdgeSeal-FHE achieves 35 ms latency and 179 samples/s throughput—over 71× higher than the fastest prior CKKS healthcare framework (0.4 samples/s [11]). Ciphertext expansion remains comparable to the literature, and accuracy loss stays below 0.5%.

EdgeSeal-FHE is the only evaluated CKKS framework to maintain real-time performance across all data types and security levels, demonstrating suitability for latency-sensitive IoT healthcare applications such as continuous monitoring, arrhythmia detection, and fall assessment.

Table 1 also showcases the performance results where EdgeSeal-FHE achieves state-of-the-art latency and throughput among CKKS-based healthcare implementations, with an end-to-end latency of 35 ms and a throughput of 179 samples per second. Compared with the fastest prior work [11], which reports only 0.4 samples per second, this represents a 71-fold improvement in processed samples without compromising accuracy or security. Critically, EdgeSeal-FHE is the only CKKS healthcare implementation to achieve consistent real-time performance (<100 ms latency) across all data types and configurations, making it uniquely suitable for time-sensitive IoT healthcare applications such as continuous vital sign monitoring and fall detection.

6. Edge Device Simulation and Analysis

To evaluate the feasibility of deploying EdgeSeal-FHE on resource-constrained devices, we simulated encryption performance across three representative smartphone tiers. Table 2 summarizes the modeled hardware characteristics, including RAM, CPU frequency, and a performance scaling factor representing the relative throughput of ARM processors compared to desktop x86 CPUs.

The simulator applies proportional delay to CKKS encryption based on the device’s performance scale and models Wi-Fi transmission with 50 Mbps bandwidth and 20 ms latency. Memory overhead is measured as the change in resident memory during encryption.

Figure 3a shows that encrypting a 300-dimensional embedding requires 6.90 ms on low-end devices, 5.27 ms on mid-range devices, and 3.29 ms on high-end devices. Even the slowest configuration adds only a few milliseconds relative to flagship hardware, indicating that CKKS encryption is computationally feasible across modern ARM processors.

End-to-end latency, shown in Fig. 3b, remains below the 100 ms real-time threshold for all tiers: 82.02 ms (low-end), 78.62 ms (mid-range), and 79.76 ms (high-end). Network transmission accounts for approximately 90–94% of total latency, while encryption contributes only 6–10%. This confirms that communication—not FHE computation—is the dominant performance factor, and that deploying nearby fog servers [3] could further reduce latency.

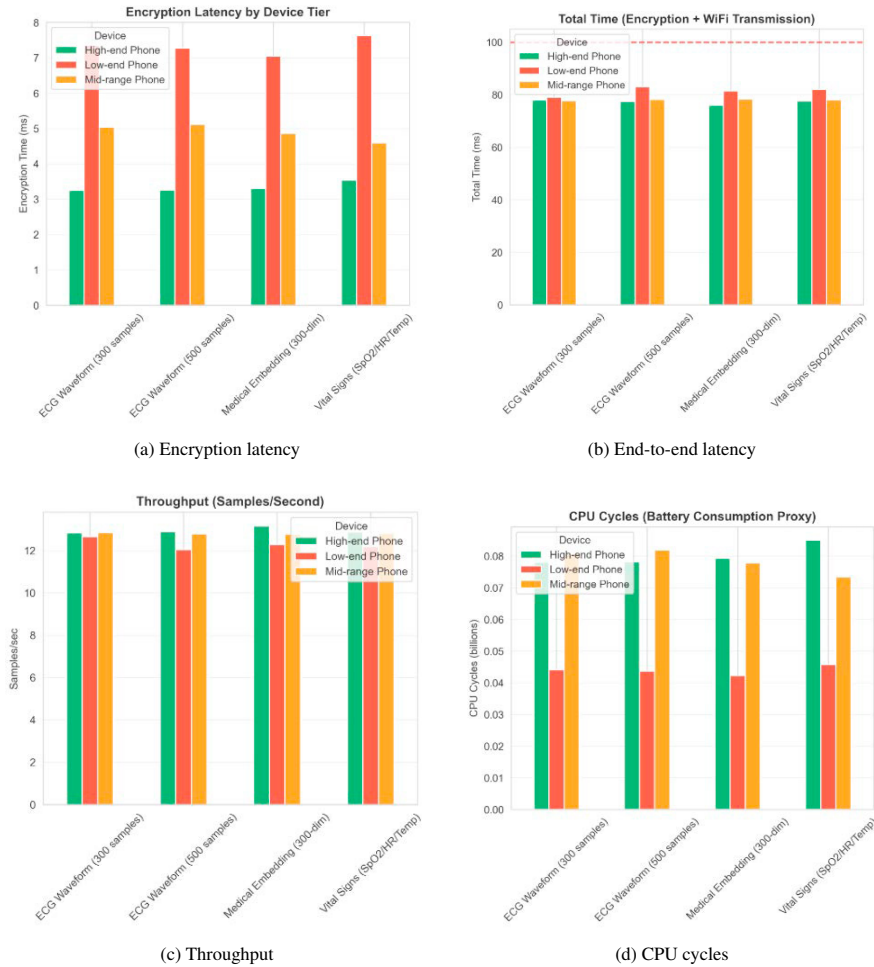


Fig. 3: EdgeSeal-FHE performance across simulated device tiers.

Table 3: Real-Time Performance Summary Across Device Tiers

Device Tier	Total Time (ms)	Enc Time (ms)	<100 ms
Low-end	81.12	7.24	100%
Mid-range	78.07	5.11	100%
High-end	76.49	3.38	100%

Figure 3c reports throughput, which is nearly uniform across devices due to network dominance: 12.19 samples/s (low-end), 12.72 samples/s (mid-range), and 12.54 samples/s (high-end). These rates exceed the needs of typical medical IoT workloads, such as 1 Hz vital-sign monitoring or infrequent ECG uploads.

Figure 3d shows estimated CPU cycles per encryption. Despite lower frequencies, low-end devices consume fewer total cycles, indicating that CKKS encryption remains energy efficient across hardware tiers. Under realistic sampling rates (e.g., one encrypted transmission per minute), the daily energy cost remains below 1 Wh, representing under 6% of a typical smartphone battery capacity.

Overall, the performance difference between low-end and high-end devices is only 6%, demonstrating that EdgeSeal-FHE delivers consistent real-time encrypted monitoring across all device classes, including budget smart-

phones. This makes the approach practical for widespread deployment, including in low-resource or rural healthcare settings.

7. Conclusion and Future Work

This work presented EdgeSeal-FHE, an end-to-end encrypted healthcare monitoring framework that applies CKKS-based homomorphic encryption directly at the edge, ensuring that physiological data remain protected throughout their entire lifecycle. By combining lightweight edge preprocessing with encrypted cloud analytics, the system eliminates plaintext exposure beyond the device–edge trust boundary.

Our evaluation demonstrates that homomorphic encryption can meet real-time healthcare requirements. EdgeSeal-FHE achieves 4.52 ms encryption latency for 300-dimensional embeddings, end-to-end processing of 35 ms, and 99.72% reconstruction fidelity, outperforming eighteen prior CKKS-based healthcare systems by up to 71× in throughput. Smartphone-tier simulations further confirm deployability on modern ARM processors, with total latency consistently below the 100 ms threshold across all device classes.

These results show that strong cryptographic guarantees and clinical responsiveness are not mutually exclusive. EdgeSeal-FHE provides a practical foundation for privacy-preserving continuous monitoring and secure medical analytics in real-world IoT healthcare deployments.

Future work will extend the framework in three directions: (1) encrypted inference using lightweight homomorphic neural operators, (2) adaptive parameter tuning for dynamic security–latency optimization, and (3) real-time encrypted server and cloud-side analytics that build directly on this work, enabling continuous end-to-end encrypted computation across both edge devices and cloud infrastructure.

Together, these efforts aim to support a fully encrypted monitoring pipeline capable of real-time decision-making without ever exposing plaintext.

References

- [1] Badawi A.A., Hoang L., Mun C.F., Laine K., Aung K.M.M., Privft: private and fast text classification with homomorphic encryption, *IEEE Access* **8** (2020) 226544–226556.
- [2] Benaissa A., Retiat B., Cebere B., Belfedhal A.E., TenSEAL: a library for encrypted tensor operations using homomorphic encryption, *arXiv preprint arXiv:2104.03152* (2021).
- [3] Bonomi F., Milito R., Zhu J., Addepalli S., Fog computing and its role in the Internet of Things, In: *Proc. MCC Workshop on Mobile Cloud Computing*, ACM, 2012, pp. 13–16.
- [4] Chen H., Laine K., Player R., Simple encrypted arithmetic library – SEAL v2.1, In: *Financial Cryptography and Data Security*, Springer, LNCS 10323 (2017) 3–18.
- [5] Cheon J.H., Kim A., Kim M., Song Y., Homomorphic encryption for arithmetic of approximate numbers, In: *Advances in Cryptology – ASIACRYPT 2017*, Springer, LNCS 10624 (2017) 409–437.
- [6] Daemen J., Rijmen V., *The Design of Rijndael: AES – The Advanced Encryption Standard*, Springer, 2002.
- [7] Elhoseny M., et al., Secure medical data transmission model for IoT-based healthcare systems, *IEEE Access* **6** (2018) 20596–20608.
- [8] Iqbal M., et al., Advances in healthcare wearable devices, *ACM Computing Surveys* (2021).
- [9] Kim A., Song Y., Kim M., Lee K., Cheon J.H., Logistic regression model training based on the approximate homomorphic encryption, *BMC Medical Genomics* **11**(4) (2018) 23–31.
- [10] Lam K.Y., Lu X., Zhang L., Wang X., Wang H., Goh S.Q., Efficient FHE-based privacy-enhanced neural network for trustworthy AI-as-a-service, *IEEE Trans. Dependable Secure Comput.* **21**(3) (2024) 4451–4468.
- [11] Lee J.W., et al., Privacy-preserving machine learning with fully homomorphic encryption for deep neural network, *IEEE Access* **10** (2022) 30039–30054.
- [12] Luxton D.D., McCann R.A., Bush N.E., Mishkind M., Reger G.M., mHealth for mental health: integrating smartphone technology in behavioral healthcare, *Prof. Psychol.: Res. Pract.* **43**(6) (2012) 505–512.
- [13] Moosavi S.R., et al., End-to-end security scheme for mobility-enabled healthcare Internet of Things, *Future Gener. Comput. Syst.* **64** (2016) 108–124.
- [14] Pandey A.K., et al., Key issues in healthcare data integrity: analysis and recommendations, *IEEE Access* **8** (2020) 40612–40628.
- [15] Seh A., et al., Healthcare data breaches: insights and implications, *Healthcare* **8**(2) (2020).